Module 7) Python
Collections, functions and Modules
Practical

1.Accessing List

Lab:

Q.1 Write a Python program to create a list with elements of multiple data types (integers, strings, floats, etc.).

Ans.

```
mixed_list = [10, "Python", 3.14, True, 25.5, "Programming"]

print(mixed_list)
```

Q.2 Write a Python program to access elements at different index positions.

Ans.

```
items = [10, "Python", 3.14, True, "Programming"]

print(items[0])

print(items[2])

print(items[-1])

print(items[-2])
```

Practical Examples:

1. Write a Python program to create a list of multiple data type elements.

Ans.

```
my_list = [10, "Python", 3.14, True, 25, "Programming"]

print(my_list)
```

2. Write a Python program to find the length of a list using the len() function.

Ans.

```
my_list = [10, "Python", 3.14, True, 25]

list_length = len(my_list)

print("Length of the list:", list_length)
```

2. List Operations

Lab:

Q.3 Write a Python program to add elements to a list using insert() and append().

Ans.

```
my_list = [10, 20, 30]

my_list.append(40)

my_list.insert(1, 15)

print(my_list)
```

Q.4 Write a Python program to remove elements from a list using pop() and remove().

Ans.

```
my_list = [10, 20, 30, 40, 50]

my_list.pop(2)

my_list.remove(40)

print(my_list)
```

Practical Examples:

3) Write a Python program to update a list using insert() and append().

Ans.

```
my_list = [10, 20, 30]

my_list.insert(1, 15)

my_list.append(40)

print(my_list)
```

4) Write a Python program to remove elements from a list using pop() and remove().

Ans.

```
my_list = [10, 20, 30, 40, 50]

my_list.pop(2)
```

```python
my_list.remove(40)

print(my_list)
```

3. Working with Lists

Lab:

Q.5 Write a Python program to iterate over a list using a for loop.

Ans.

```python
my_list = [10, 20, 30, 40, 50]

for item in my_list:
    print(item)
```

Q.6 Write a Python program to sort a list using both sort() and sorted().

Ans.

```python
my_list = [50, 10, 30, 20, 40]

my_list.sort()

print("List after sort():", my_list)

my_list2 = [50, 10, 30, 20, 40]

sorted_list = sorted(my_list2)

print("List after sorted():", sorted_list)
```

Practical Examples:

5) Write a Python program to iterate through a list and print each element.

Ans.

```
my_list = [10, 20, 30, 40, 50]

for element in my_list:
    print(element)
```

6) Write a Python program to insert elements into an empty list using a for loop and append().

Ans.

```
my_list = []

for i in range(1, 6):
    my_list.append(i)

print(my_list)
```

4. Tuple

Lab:

Q.7 Write a Python program to create a tuple with multiple data types.

Ans.

```
my_tuple = (10, "Python", 3.14, True, "Programming")

print(my_tuple)
```

Q.8 Write a Python program to concatenate two tuples.

Ans.

```
tuple1 = (1, 2, 3)
tuple2 = ("a", "b", "c")

result_tuple = tuple1 + tuple2

print(result_tuple)
```

Practical Examples:

7) Write a Python program to convert a list into a tuple.

Ans.

```
my_list = [10, 20, 30, 40]

my_tuple = tuple(my_list)

print(my_tuple)
```

8) Write a Python program to create a tuple with multiple data types.

Ans.

```
mixed_tuple = ("Apple", 42, 3.14, True, [1, 2])

print("Mixed tuple:", mixed_tuple)

print("Tuple length:", len(mixed_tuple))
```

9) Write a Python program to concatenate two tuples into one.

Ans.

```
tuple1 = (1, 2, 3)

tuple2 = (4, 5, 6)

result_tuple = tuple1 + tuple2

print("Tuple 1:", tuple1)

print("Tuple 2:", tuple2)

print("Concatenated:", result_tuple)
```

10) Write a Python program to access the value of the first index in a tuple.4

Ans.

```
my_tuple = (10, 20, 30, 40, 50)

first_element = my_tuple[0]

print("Original tuple:", my_tuple)

print("First element (index 0):", first_element)
```

5. Accessing Tuples

Lab:

Q.9 Accessing tuple elements using positive and negative indexing.

Ans.

```
my_tuple = (10, 20, 30, 40, 50, 60, 70)

slice_result = my_tuple[1:5]

print("Original tuple:", my_tuple)

print("Elements from index 1 to 5:", slice_result)
```

Q.10 Write a Python program to access alternate values between index 1 and 5 in a tuple.

Ans.

```
my_tuple = (0, 10, 20, 30, 40, 50, 60, 70)

alternate_slice = my_tuple[1:5:2]

print("Original tuple:", my_tuple)

print("Alternate values (indices 1,3):", alternate_slice)
```

Practical Examples:

11) Write a Python program to access values between index 1 and 5 in a tuple.

Ans.

```
my_tuple = (0, 10, 20, 30, 40, 50, 60, 70)

result = my_tuple[1:5]

print("Original tuple:", my_tuple)

print("Slice [1:5]:", result)
```

12) Write a Python program to access the value from the last index in a tuple.

Ans.

```
my_tuple = (10, 20, 30, 40, 50)

last_element = my_tuple[-1]

print("Original tuple:", my_tuple)
print("Last element:", last_element)
```

6. Dictionaries

Lab:

Q.11 Write a Python program to create a dictionary with 6 key-value pairs.

Ans.

```
my_dict = {
    "name": "Rohit",
    "age": 22,
```

```python
    "course": "B.tech",
    "language": "Python",
    "city": "Ahmedabad",
    "is_student": True
}

print(my_dict)
```

Q.12 Write a Python program to access values using dictionary keys.

Ans.

```python
student = {
    "name": "Rohit",
    "age": 22,
    "grade": "A",
    "city": "Mumbai"
}
print("Name:", student["name"])

print("Age:", student["age"])

print("Grade:", student["grade"])
```

Practical Examples:

13) Write a Python program to create a dictionary of 6 key-value pairs.

Ans.

```
person_data = {
    "name": "Mahi",
    "age": 28,
    "salary": 65000.50,
    "is_married": False,
    "city": "Ahmedabad",
    "skills": ["Python", "JavaScript", "SQL"]
}
print("Person Dictionary:", person_data)
print("Total pairs:", len(person_data))

print("\nKey-value pairs:")
for key, value in person_data.items():
    print(f"{key}: {value}")
```

14) Write a Python program to access values using keys from a dictionary.

Ans.

```
car = {
    "brand": "Toyota",
    "model": "Camry",
    "year": 2023,
    "price": 28000
}
```

```python
print("Brand:", car["brand"])
print("Model:", car["model"])
print("Year:", car["year"])
print("Price:", car["price"])
```

7. Working with Dictionaries

Lab:

Q.13 Write a Python program to update a value in a dictionary.

Ans.

```python
employee = {
    "name": "Mahi",
    "salary": 50000,
    "department": "IT"
}
print("Before update:", employee)

employee["salary"] = 65000

print("After salary update:", employee)
```

Q.14 Write a Python program to merge two lists into one dictionary using a loop.

Ans.

```python
keys = ["name", "age", "city", "job", "salary", "active"]
values = ["Alice", 25, "London", "Engineer", 75000, True]
my_dict = {}
for key, value in zip(keys, values):
    my_dict[key] = value
print("Keys list:", keys)
```

```
print("Values list:", values)
print("Merged dictionary:", my_dict)
print("Dictionary length:", len(my_dict))
```

Practical Examples:

15) Write a Python program to update a value at a particular key in a dictionary.

Ans.

```
book = {
    "title": "Python Programming",
    "price": 29.99,
    "pages": 450,
    "author": "John"
}
print("Before update:", book)
book["price"] = 34.99
print("After price update:", book)
print("New price:", book["price"])
```

16) Write a Python program to separate keys and values from a dictionary using keys() and values() methods.

Ans.

```
employee = {
    "name": "sky",
    "age": 28,
    "department": "IT",
    "salary": 75000,
    "city": "Mumbai",
```

```
    "active": True
}
keys_list = list(employee.keys())
values_list = list(employee.values())
print("Original dictionary:", employee)
print("Keys:", keys_list)
print("Values:", values_list)
```

17) Write a Python program to convert two lists into one dictionary using a for loop.

Ans.

```
keys = ['name', 'age', 'city']
values = ['Rohit', 25, 'Mumbai']
my_dict = {}

for i in range(len(keys)):
    my_dict[keys[i]] = values[i]

print("Dictionary from lists:", my_dict)
```

18) Write a Python program to count how many times each character appears in a string.

Ans.

```
text = "python programming"
char_count = {}
for char in text:
    if char != " ":
        if char in char_count:
```

```
            char_count[char] += 1
        else:
            char_count[char] = 1
print(char_count)
```

8. Functions

Lab:

Q.15 Write a Python program to create a function that takes a string as input and prints it.

Ans

```
def print_string(text):
    print(text)


print_string("Hello, Python!")
```

Q.16 Write a Python program to create a calculator using functions.

Ans.

```
def add(x, y):
    return x + y


def subtract(x, y):
    return x - y


def multiply(x, y):
    return x * y


def divide(x, y):
    if y == 0:
        return "Error: Division by zero!"
```

```
    return x / y

def calculator():
    print("=== Simple Calculator ===")
    print("Select operation:")
    print("1. Add (+)")
    print("2. Subtract (-)")
    print("3. Multiply (*)")
    print("4. Divide (/)")
    print("5. Exit")

    while True:
        choice = input("\nEnter choice (1/2/3/4/5): ")

        if choice == '5':
            print("Goodbye!")
            break

        if choice in ['1', '2', '3', '4']:
            try:
                num1 = float(input("Enter first number: "))
                num2 = float(input("Enter second number: "))

                if choice == '1':
                    result = add(num1, num2)
                elif choice == '2':
                    result = subtract(num1, num2)
```

```
            elif choice == '3':
                result = multiply(num1, num2)
            elif choice == '4':
                result = divide(num1, num2)

            print(f"Result: {num1} {['+', '-', '*', '/'][int(choice)-1]} {num2} = 
{result}")
        except ValueError:
            print("Invalid input! Please enter numbers.")
    else:
        print("Invalid choice! Please select 1-5.")

calculator()
```

Practical Examples:

19) Write a Python program to print a string using a function.

Ans.

```
def print_message():
    print("Welcome to Python Programming")

print_message()
```

20) Write a Python program to create a parameterized function that takes two arguments and prints their sum.

Ans.

```
def calculate_sum(a, b):
    sum_result = a + b
    print(f"The sum of {a} and {b} is: {sum_result}")
calculate_sum(5, 3)
```

```
calculate_sum(10.5, 7.2)
```

21) Write a Python program to create a lambda function with one expression.

Ans.

```
square = lambda x: x ** 2

print("Lambda function result:", square(5))

print("Lambda function result:", square(7))
```

22) Write a Python program to create a lambda function with two expressions.

Ans.

```
add_numbers = lambda x, y: x + y


print("Sum using lambda:", add_numbers(8, 5))

print("Sum using lambda:", add_numbers(12, 7))
```

9. Modules

Lab:

Q.17 Write a Python program to import the math module and use functions like sqrt(), ceil(), floor().

Ans.

```
import math

print("Square root of 16:", math.sqrt(16))


print("Ceil of 4.7:", math.ceil(4.7))

print("Ceil of -2.3:", math.ceil(-2.3))


print("Floor of 4.7:", math.floor(4.7))

print("Floor of -2.3:", math.floor(-2.3))
```

Q.18 Write a Python program to generate random numbers using the random module.

Ans.

```
import random
print("Random integer (1-100):", random.randint(1, 100))


print("Random float (0-1):", random.random())
print("Random integer (5-20):", random.randrange(5, 20))
options = ['apple', 'banana', 'orange']
print("Random choice:", random.choice(options))
```

Practical Examples:

23) Write a Python program to demonstrate the use of functions from the math module.

Ans.

```
import math


number = 16
value = 7.3
print("Square root:", math.sqrt(number))
print("Power:", math.pow(2, 3))
print("Ceil value:", math.ceil(value))
print("Floor value:", math.floor(value))
print("Factorial:", math.factorial(5))
```

24) Write a Python program to generate random numbers between 1 and 100 using the random module.

Ans.

random module

```
import random
random_number = random.randint(1, 100)
print(random_number)
```